

Multiple Choice (40 marks)

1. Why might you choose to use a silicone catheter (as of 2020 medical knowledge)?
 - (a) Because the patient does not like the others.
 - (b) Because the patient is allergic to latex.
 - (c) Because there are no other kinds of catheter at your disposal.
 - (d) Because it is cheaper.
2. The sales tax that you pay at the grocery store is commonly labeled a
 - (a) progressive tax.
 - (b) regressive tax.
 - (c) proportional tax.
 - (d) excise tax.
3. Cancer is best described as
 - (a) a type of genetic engineering
 - (b) the merging of sex cells
 - (c) abnormal cell division
 - (d) tissue that stops growing
4. You have been asked to write a response paper to LeVay's research on brain differences between heterosexuals and homosexuals. Which are plausible criticisms...
 - (a) Sample size was relatively small.
 - (b) Gay and straight groups were not equivalent in terms of the number of deaths due to AIDS.
 - (c) Lesbians were omitted from the study.
 - (d) all of the above
5. Which of the following is/are true?
 - (a) Titan is the only outer solar system moon with a thick atmosphere
 - (b) Titan is the only outer solar system moon with evidence for recent geologic activity
 - (c) Titan's atmosphere is composed mostly of hydrocarbons
 - (d) A and D
6. Aesthetics deals with objects that are_____.

- (a) essential to our existence
 - (b) unimportant to most people
 - (c) not essential to our existence
 - (d) rarely viewed
7. Which is an example of a single-celled organism?
- (a) amoeba
 - (b) insect
 - (c) fish
 - (d) worm
8. Each story of a building is 14.3 feet tall. If the height of the building is 185.9 feet, how many floors does the building have?
- (a) 10
 - (b) 13
 - (c) 17
 - (d) 9
9. Where would a hare live?
- (a) in the zoo for its safety from sharks
 - (b) a field for sustenance
 - (c) space for its low gravity
 - (d) a desert for its high temperatures
10. At the birthday party of your best friend, you see Skylar help himself to a second piece of cake. For this individual, it must be the case that
- (a) the marginal benefit of the second piece of cake is less than the marginal cost.
 - (b) the total benefit received from eating cake is falling.
 - (c) the ratio of marginal benefit over marginal cost is less than one.
 - (d) the marginal benefit of the second piece of cake is greater than the marginal cost.
11. What is the meaning of the term "heteroscedasticity"?
- (a) The variance of the errors is not constant
 - (b) The variance of the dependent variable is not constant
 - (c) The errors are not linearly independent of one another
 - (d) The errors have non-zero mean

12. Refraction
- (a) makes objects appear to be twice their size
 - (b) moves heat energy thru sound waves
 - (c) is the phenomenon involved in mirrors
 - (d) makes a straw appear broken when in a cup of water
13. The polynomial $x^3 + 2x^2 + 2x + 1$ can be factored into linear factors in $\mathbb{Z}_7[x]$. Find this factorization.
- (a) $(x - 2)(x + 2)(x - 1)$
 - (b) $(x + 1)(x + 4)(x - 2)$
 - (c) $(x + 1)(x - 4)(x - 2)$
 - (d) $(x - 1)(x - 4)(x - 2)$
14. Which instrument would be best to measure the volume of a liquid?
- (a) a pan balance
 - (b) a stopwatch
 - (c) a thermometer
 - (d) a graduated cylinder
15. Which concept supports the endosymbiotic origin of complex, eukaryotic double-membrane structures?
- (a) Mitochondria and chloroplasts arise from preexisting mitochondria and chloroplasts.
 - (b) Mitochondria and chloroplasts arise from preexisting endomembrane systems.
 - (c) Mitochondria and chloroplasts are synthesized in the nucleus.
 - (d) Mitochondria and chloroplasts are synthesized in ribosomes.
16. The major beneficiary of Medicaid in the U.S. is:
- (a) Single men
 - (b) The elderly
 - (c) Women and children
 - (d) Undocumented residents
17. Which of the following best describes an anthropologist who studies the human skeleton in order to solve crimes?
- (a) forensic anthropologist
 - (b) historical archaeologist

- (c) paleoanthropologist
 - (d) linguistic anthropologist
18. As the molecules of water begin to slow down they go into a phase where they are no longer able to easily move past each other. The water is passing through...
- (a) remain a mixture.
 - (b) remain a compound.
 - (c) become an element.
 - (d) become a compound.
19. To look at an eclipse you might be able to use
- (a) nothing
 - (b) telescope
 - (c) Plain old sunglasses
 - (d) a box
20. Which of the following must exist in a population before natural selection can act upon that population?
- (a) Genetic variation among individuals
 - (b) Variation among individuals caused by environmental factors
 - (c) Sexual reproduction
 - (d) B and C only

End of Paper